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Inventor(s)

Fowler; Simon et al.

TWO DIMENSIONAL BREW SELECTION INTERFACE FOR COFFEE MAKER WITH SUGGESTION CAPABILITY

Abstract

A two dimensional brew selection interface is used to present suggestions for configuring a plurality of brewing parameters used in a coffee brewing operation. The two dimensional brew selection interface may include a plurality of positions distributed across multiple dimensions, with each position mapped to a set of values for multiple brewing parameters, and one or more indicators associated with suggested combinations of brewing parameters may be presented on the two dimensional brew selection interface, e.g., based at least in part upon a user's history and/or feedback.

Inventors: Fowler; Simon (Louisville, KY), Hunter; Matthew (Louisville, KY), Persaud; Ryan (Louisville, KY)

Applicant: Midea Group Co., Ltd. (Foshan, CN)

Family ID: 1000007724911

Assignee: Midea Group Co., Ltd. (Foshan, CN)

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Background/Summary

BACKGROUND

[0001] Coffee is a beverage made from roasted coffee beans, and has long been one of the most popular drinks in the world. Numerous devices and processes for making coffee have been developed, with most incorporating some variation on bringing water into contact with ground coffee beans to extract water soluble components from the ground coffee beans that give coffee its distinctive taste. However, roasted coffee beans can have numerous soluble components that impart different types of flavors that can be pleasing or displeasing to different palates, so variations in brewing processes can significantly impact the flavor of the resulting coffee.

[0002] Two primary types of coffee brewing devices in common use today are generally referred to as espresso machines and drip coffee makers. Espresso machines force hot pressurized water relatively quickly through packed and finely ground coffee beans to generate a highly concentrated, but low volume type of coffee generally referred to as espresso. Drip coffee makers, on the other hand, rely primarily on gravity rather than pressure, and drip hot water onto a basket of more coarsely ground coffee beans to allow the water to seep through the coffee grounds to extract the flavor-producing components therefrom. Drip coffee makers also generally incorporate a disposable or reusable filter in the basket such that the coffee grounds are retained in the basket while the coffee is fed by gravity into a cup or larger container disposed below the basket.

[0003] Particularly with drip coffee makers, a number of factors can greatly impact a coffee flavor profile, including, for example, the amounts of ground coffee and water that are used, the grind size used to grind the coffee beans, the temperature of the water, and the amount of time the water contacts the ground coffee, among others. While some of these factors, e.g., the grind size and the amount of ground coffee, are often controllable by a consumer, other factors may not be readily controllable. Furthermore, different coffee beans can generate differently-tasting coffees even under identical brewing conditions, and despite industry efforts to characterize a “golden cup” brewing standard, different consumers can still have different taste preferences, so what one consumer considers to be a perfect cup of coffee may differ substantially from the opinions of other consumers.

[0004] Efforts have been made to improve the reproducibility of drip coffee makers and thereby minimize the variability of some of the factors that can impact a coffee flavor profile, e.g., by tightly controlling water temperature so that all coffee is brewed at a consistent temperature, so that fewer factors need to be considered when attempting to brew a good tasting coffee. Nonetheless, for many consumers, they are either not sufficiently experienced or not particularly interested in putting in the effort to determine the different brewing parameters needed to brew their own “perfect” cup of coffee. Many consumers value simplicity and speed over taste, and will often reluctantly accept subpar tasting coffee as long as it is quick and easy to brew. On the other hand, this can be to the detriment of the manufacturer of the drip coffee maker, as the consumer may simply blame the product for brewing mediocre coffee, despite the fact that the brewing parameters that were used were not optimal.

[0005] Therefore, a significant need exists in the art for a manner of simplifying the configuration of a drip coffee maker to enable consumers to better brew coffee to their own personal liking.

SUMMARY

[0006] The herein-described embodiments address these and other problems associated with the art by utilizing a two dimensional brew selection interface to present suggestions for configuring a plurality of brewing parameters used in a coffee brewing operation. The two dimensional brew selection interface may include a plurality of positions distributed across multiple dimensions, with each position mapped to a set of values for multiple brewing parameters, and one or more

indicators associated with suggested combinations of brewing parameters may be presented on the two dimensional brew selection interface, e.g., based at least in part upon a user's history and/or feedback.

[0007] Therefore, consistent with one aspect of the invention, an apparatus may be provided including one or more processors and memory operably coupled with the one or more processors, the memory storing instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by presenting a two dimensional brew selection interface to a user, the two dimensional brew selection interface including a plurality of positions distributed across first and second dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation, presenting on the two dimensional brew selection interface a plurality of historical indicators respectively associated with a plurality of prior coffee brewing operations, each of the plurality of historical indicators having a respective position in the two dimensional brew selection interface corresponding to a respective set of values for the plurality of brewing parameters used when performing the associated prior coffee brewing operation, determining, based on at least a portion of the plurality of prior coffee brewing operations, a suggested set of values for the plurality of brewing parameters, and presenting on the two dimensional brew selection interface a suggested indicator associated with a suggested coffee brewing operation to be performed, the suggested indicator having a position in the two dimensional brew selection interface corresponding to the suggested set of values for the plurality of brewing parameters.

[0008] In some embodiments, a first historical indicator of the plurality of historical indicators is a rating indicator that indicates a user rating associated with a first prior coffee brewing operation of the plurality of prior coffee brewing operations. Also, in some embodiments, the one or more processors are configured to present the first historical indicator with a size that is based at least in part on the user rating associated with the first prior coffee brewing operation.

[0009] Further, in some embodiments, the one or more processors are configured to determine the suggested set of values based at least in part on the user rating of the first prior coffee brewing operation. In some embodiments, a first historical indicator of the plurality of historical indicators is a favorite indicator that indicates that a user favorited an associated first prior coffee brewing operation of the plurality of prior coffee brewing operations. In addition, in some embodiments, a second historical indicator of the plurality of historical indicators is a rating indicator that indicates a user rating associated with a second prior coffee brewing operation of the plurality of prior coffee brewing operations.

[0010] In some embodiments, the one or more processors are configured to determine the suggested set of values based at least in part on the user favoriting the first prior coffee brewing operation. In addition, in some embodiments, the one or more processors are configured to update the position of the favorite indicator to that of the suggested indicator in response to user input received subsequent to a coffee brewing operation performed using the suggested set of values.

[0011] Moreover, in some embodiments, the one or more processors are further configured to receive user input from the user selecting the suggested coffee brewing operation, and configure the plurality of brewing parameters for the coffee brewing operation to use the suggested set of values for the suggested coffee brewing operation.

[0012] In some embodiments, the apparatus is a mobile computing device, the one or more processors present the two dimensional brew selection interface on and receive the user input from a touch screen display of the mobile computing device, and the one or more processors configure the plurality of brewing parameters for the coffee brewing operation by communicating data to a coffee maker to cause the coffee maker to use the suggested set of values corresponding to the suggested coffee brewing operation. Moreover, in some embodiments, the apparatus is a drip coffee maker including a brew basket, a heating element configured to heat water supplied to the brew basket, and a flow control device configured to control water flow through the brew basket, the one

or more processors present the two dimensional brew selection interface on and receive the user input from a touch screen display of the drip coffee maker, and the one or more processors further configure the coffee brewing operation by controlling at least one of the heating element and the flow control device during the coffee brewing operation based upon at least a portion of the values in the suggested set of values corresponding to the suggested coffee brewing operation.

[0013] In some embodiments, the plurality of brewing parameters includes a water temperature, a water pressure, a water flow rate, a valve position, a water amount, a coffee amount, a grind size, a bean type, a roast level, a brewing ratio, and/or a water/coffee contact time. In addition, in some embodiments, the plurality of brewing parameters includes a coffee amount parameter, and the one or more processors are configured to calculate a value of the coffee amount parameter based at least in part on the position of the suggested indicator in the two dimensional brew selection interface and prompt the user to add an amount of coffee associated with the calculated value of the coffee amount parameter. In some embodiments, the plurality of brewing parameters includes a coffee amount parameter, and the one or more processors are configured to calculate a value of the coffee amount parameter based at least in part on the position of the suggested indicator in the two dimensional brew selection interface and control a coffee grinder to grind an amount of coffee associated with the calculated value of the coffee amount parameter.

[0014] Moreover, in some embodiments, the user input is first user input, and the one or more processors are further configured to rate the coffee brewing operation after completion of the coffee brewing operation in response to second user input. Also, in some embodiments, the first dimension corresponds to strength and the second dimension corresponds to development, and the strength is based upon concentration of solubles and the development is based upon yield of solubles. In some embodiments, the plurality of prior coffee brewing operations is a first plurality of prior coffee brewing operations, and the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters further based on user profile data associated with a user, a second plurality of prior coffee brewing operations performed on a plurality of coffee makers by a plurality of users, time, date, or location data, coffee bean data, or coffee drink type data.

[0015] In addition, in some embodiments, the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters using a machine learning model trained based upon a plurality of historical brewing operations. Also, in some embodiments, the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters using a mathematical distance optimization algorithm that optimizes the position of the suggested indicator in the two dimensional brew selection interface based on at least a portion of the plurality of prior coffee brewing operations. Moreover, in some embodiments, the mathematical distance optimization algorithm is configured to determine the optimized position of the suggested indicator in the two dimensional brew selection interface using the respective positions of at least a portion of the plurality of historical indicators.

[0016] Further, in some embodiments, the one or more processors are further configured to determine a coffee bean recommendation based on at least a portion of the plurality of prior coffee brewing operations, and present the coffee bean recommendation to the user.

[0017] Consistent with another aspect of the invention, an apparatus may be provided including one or more processors and memory operably coupled with the one or more processors, the memory storing instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by presenting an N-dimensional brew selection interface to a user, the N-dimensional brew selection interface including a plurality of positions distributed across N dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation, presenting on the N-dimensional brew selection interface a plurality of historical indicators respectively associated with a plurality of prior coffee brewing operations, each of the

plurality of historical indicators having a respective position in the N-dimensional brew selection interface corresponding to a respective set of values for the plurality of brewing parameters used when performing the associated prior coffee brewing operation, determining, based on at least a portion of the plurality of prior coffee brewing operations, a suggested set of values for the plurality of brewing parameters, and presenting on the N-dimensional brew selection interface a suggested indicator associated with a suggested coffee brewing operation to be performed, the suggested indicator having a position in the N-dimensional brew selection interface corresponding to the suggested set of values for the plurality of brewing parameters.

[0018] Consistent with yet another aspect of the invention, an apparatus may be provided including one or more processors and memory operably coupled with the one or more processors, the memory storing instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by representing a plurality of prior coffee brewing operations in an N-dimensional space, the N-dimensional space including a plurality of positions distributed across N dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation, and with each of the prior coffee brewing operations having an associated user-assigned rating, determining, based on at least a portion of the plurality of prior coffee brewing operations and the user-assigned ratings associated therewith, a suggested position in the N-dimensional space, and suggesting a suggested coffee brewing operation to be performed using the set of values for the plurality of brewing parameters corresponding to the suggested position in the N-dimensional space.

[0019] Other embodiments may include various methods for making and/or using any of the aforementioned constructions.

[0020] These and other advantages and features, which characterize the invention, are set forth in the claims annexed hereto and forming a further part hereof. However, for a better understanding of the invention, and of the advantages and objectives attained through its use, reference should be made to the Drawings, and to the accompanying descriptive matter, in which there is described example embodiments of the invention. This summary is merely provided to introduce a selection of concepts that are further described below in the detailed description, and is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 is a front elevational view of a coffee maker consistent with some embodiments of the invention.

[0022] FIG. 2 is a block diagram of an example control system for the coffee maker of FIG. 1.

[0023] FIG. 3 is a flowchart illustrating an example operational sequence for controlling the coffee maker of FIGS. 1 and 2 using a two dimensional brew selection interface.

[0024] FIGS. 4-20 are representative user interface displays that may be displayed by the coffee maker of FIGS. 1-2.

[0025] FIG. 21 is a flowchart illustrating an example operational sequence for controlling the coffee maker of FIGS. 1 and 2 using a scanned barcode from coffee bean packaging.

[0026] FIG. 22 is a flowchart illustrating an example operational sequence for controlling the coffee maker of FIGS. 1 and 2 to perform a suggested coffee brewing operation.

[0027] FIG. 23 is a representative user interface display that presents a suggested indicator for a suggested coffee brewing operation capable of being performed by the coffee maker of FIGS. 1-2.

[0028] FIG. 24 is a block diagram illustrating an example embodiment of a suggestion

determination operation capable of being performed by the coffee maker of FIGS. 1-2.

[0029] FIG. 25 is a representative user interface display that presents a coffee bean recommendation generated by the suggestion determination operation of FIG. 24.

[0030] FIG. 26 is a block diagram illustrating an example coffee brewing suggestion system incorporating a coffee brewing suggestion machine learning model suitable for use in performing a suggestion determination operation with the coffee maker of FIGS. 1-2.

[0031] FIG. 27 is a block diagram illustrating a training system for training the coffee brewing suggestion machine learning model of FIG. 26.

DETAILED DESCRIPTION

[0032] Now turning to the drawings, wherein like parts are denoted by like numbers throughout the several views, FIG. 1 illustrates a drip coffee maker 10 consistent with some embodiments of the invention. Drip coffee maker 10 may include a base or housing 12 within which may be included various components utilized in the brewing of coffee, e.g., a heating element 14, and in some instances, a water pump 16, for heating and conveying water from a water reservoir 18 to a water dispenser 20 disposed above a brew basket 22 within which may be disposed a quantity of ground coffee, and in many instances, a disposable or reusable filter. Brew basket 22 in some embodiments may be removable and/or may incorporate a lid or cover providing access to the brew basket 22 when adding ground coffee and a filter prior to brewing and/or when removing the ground coffee and filter after brewing. In some embodiments, a filter support 24 may be used to support a filter, and the coffee grounds contained therein, during a drip coffee brewing operation. Water dispenser 20 in some embodiments may include a shower head or other suitable mechanism for dispensing heated water evenly across the top surface of the ground coffee disposed in brew basket 22 during brewing.

[0033] Disposed beneath brew basket 22 is a coffee dispenser 26 that collects the brewed coffee that passes through the filter in brew basket 22 to dispense the brewed coffee into a container via gravity feed, e.g., a carafe 28, which in some instances may be a glass carafe or a thermal carafe, although brewing may be permitted into other types of containers, e.g., mugs or cups. Coffee dispenser 26 may, in some instances, also include an automatic shut-off valve that shuts off dispensing during a brewing operation if the carafe is removed by a consumer. In some instances, a warming element 30 may also be provided in order to maintain a desirable coffee temperature after brewing.

[0034] A user interface 32, e.g., including a graphical and/or touchscreen display 34, one or more physical controls 36 (e.g., buttons, knobs, sliders, etc.) and/or one or more indicators 38 (e.g., lights, alphanumeric displays, etc.), may also be disposed on base 12 to enable a user to interact with the drip coffee maker 10.

[0035] In some embodiments, a coffee grinder 40 may also be integrated with and/or interfaced with drip coffee maker 10. Coffee grinder 40, for example, may be integrally disposed in base or housing 12 in some embodiments, while in other embodiments, coffee grinder 40 may be removably coupled to or positionable proximate base or housing 12 and interfaced through one or more wires, one or more contacts, or a wireless network to enable commands and/or data to be communicated therebetween. Coffee grinder 40 may also be powered by drip coffee maker 10 or may be separately powered in various embodiments. In still other embodiments, no coffee grinder may be used, or to the extent a consumer grinds coffee for use with drip coffee maker 10, he or she does so using a completely separate coffee grinder.

[0036] Coffee grinder 40 may include a bin or hopper 42 positioned over a burr or blade grinding mechanism 44 that outputs ground coffee from a chute 46 into a container 48. Bin or hopper 42 may also include a lid or cover 50 for allowing for external access to add roasted coffee beans to the bin or hopper. Grinding mechanism 44 may be configured to support variable grind sizes in some embodiments, e.g., through manual or electronically-controlled adjustments to the grinding mechanism. In addition, in some embodiments, a scale 52 may be disposed in coffee grinder 40 to

measure the weight of the coffee dispensed through chute **46**. User interaction with coffee grinder **40** may be provided through a dedicated interface **54** in some embodiments, while in other embodiments, user interface **32** of drip coffee maker **10** may also control coffee grinder **40**. [0037] With further reference to FIG. **2**, drip coffee maker **10** may be under the control of a controller **60** that receives inputs from a number of components and drives a number of components in response thereto. Controller **60** may, for example, include one or more processors **62** and a memory **64** within which may be stored program code or instructions for execution by the one or more processors **62**. The memory may be embedded in controller **60**, but may also be considered to include volatile and/or non-volatile memories, cache memories, flash memories, programmable read-only memories, read-only memories, etc., as well as memory storage physically located elsewhere from controller **60**, e.g., in a mass storage device or on a remote computer interfaced with controller **60**. Controller **60** may also be implemented as a microcontroller in some embodiments, and as such these terms are used interchangeably herein. Controller **60** may also include discrete circuit logic in some embodiments, e.g., including passive and/or active circuit components.

[0038] As shown in FIG. **2**, controller **60** may be interfaced with various components, including the aforementioned heating element **14**, pump **16**, warming element **30**, user interface **32**, and coffee grinder **40**, as well as one or more electronically-controlled valves **66** and one or more sensors **68**.

[0039] In some embodiments, for example, electronically-controlled valves **66** may be disposed upstream and/or downstream of brew basket **22** to effectively control the amount of time that water contacts the ground coffee in the brew basket. An upstream valve **66**, for example, may be used to control the flow of heated water into the brew basket **22**. A downstream valve **66** may be used to control the flow of coffee out of brew basket **22** into carafe **28** or another container, and in some embodiments, such a valve may also be fully closeable to prevent coffee from dripping from the brew basket whenever carafe **28** is removed. A downstream valve **66** may also, in some embodiments, enable alternate types of brewing processes, e.g., cold brewing processes where unheated water is steeped in coffee grounds for a relatively long duration. While in some embodiments upstream and/or downstream valves **66** may be used, in other embodiments, no electronically-controllable valves may be used and/or one or both of valves **66** may be mechanically controlled. For the purposes of this disclosure, each of pump **16** and valves **66** may be considered to be a flow control device insofar as each may be used in the control of water flow through the brew basket (i.e., either into the brew basket, out of the brew basket, or both).

[0040] A wide variety of different types of sensors **68** may also be used, including for example, temperature sensors, fluid pressure sensors, fluid level sensors, flowmeters, presence sensors (e.g., to detect the proper positioning of a carafe, a brew basket, a water reservoir, a grinder hopper, etc.), position sensors, weight sensors, etc. Other sensors that may be suitable in monitoring the status of and otherwise controlling the operation of drip coffee maker **10** will be appreciated by those of ordinary skill having the benefit of the instant disclosure.

[0041] As noted above, where coffee grinder **40** is implemented as an internal coffee grinder, controller **60** may be configured to control various components of the coffee grinder directly, e.g., a motor **70** that drives the grinding process (e.g., by rotating one burr cutting element relative to another burr cutting element), and where electronic control of grind size is supported, a size control **72**. Size control in other embodiments may be a manual adjustment.

[0042] Alternatively, where drip coffee maker **10** is interfaced with an external coffee grinder (e.g., external coffee grinder **74** of FIG. **2**, shown including a motor **76**, optional size control **78**, and one or more sensors **80**), controller **60** may communicate with the external coffee grinder over a network **82** through a network interface **84** to instruct the external coffee grinder to grind coffee, e.g., to grind a desired amount (e.g., based on weight or duration) and/or to grind at a specified grind size.

[0043] Network interface **84**, for example, may represent one or more network interfaces suitable

for interfacing with external devices via wired and/or wireless networks such as Ethernet, Bluetooth, NFC, cellular and other suitable networks. It may also be desirable, for example, to interface with one or more user devices **86**, e.g., a consumer's mobile phone, which may include one or more processors **88**, a memory **90** and a user interface **92**) to enable a customer to control drip coffee maker **10** through the user device **86**. It may also be desirable to interface with one or more remote services **94**, e.g., to obtain firmware updates, to access remote databases with recipes, coffee bean information, to persist user preferences, to provide maintenance or diagnostic functionality, etc. Moreover, in some embodiments, at least a portion of controller **60** may be implemented externally, e.g., within a mobile device, a cloud computing environment, etc., such that at least a portion of the functionality described herein is implemented within the portion of the controller that is externally implemented.

[0044] In some embodiments, controller **60** may operate under the control of an operating system and may execute or otherwise rely upon various computer software applications, components, programs, objects, modules, data structures, etc. In addition, controller **60** may also incorporate hardware logic to implement some or all of the functionality disclosed herein. Further, in some embodiments, the operational sequences performed by controller **60** to implement the embodiments disclosed herein may be implemented using program code including one or more instructions that are resident at various times in various memory and storage devices, and that, when read and executed by one or more hardware-based processors, perform the operations embodying desired functionality. Moreover, in some embodiments, such program code may be distributed as a program product in a variety of forms, and that the invention applies equally regardless of the particular type of computer readable media used to actually carry out the distribution, including, for example, non-transitory computer readable storage media. In addition, it will be appreciated that the various operations described herein may be combined, split, reordered, reversed, varied, omitted, parallelized and/or supplemented with other techniques known in the art, and therefore, the invention is not limited to the particular sequences of operations described herein.

[0045] It will be appreciated that drip coffee maker **10** may be configured in a wide variety of other manners and may omit one or more of the components discussed above and/or may incorporate other components not explicitly discussed above. For example, as illustrated in FIG. **2**, it may be desirable in some embodiments to utilize a bar code or other scanner **96** to enable coffee bean or ground coffee bar codes to be scanned to identify a particular coffee bean or ground coffee being used for a coffee brewing operation and thereby enable the coffee brewing operation to be specifically tailored for a particular coffee bean or ground coffee. Such functionality could also be supported in user device **86**, e.g., using a camera of the user device to capture an image of a bar code.

[0046] Numerous additional variations and modifications to the drip coffee maker illustrated in FIGS. **1-2** will be apparent to one of ordinary skill in the art, as will become apparent from the description below. In addition, it will be appreciated that at least some of the concepts discussed herein may be utilized in connection with various types of coffee makers other than drip coffee makers and/or in connection with other types of coffee brewing processes, including, but not limited to, espresso, cold brew, French press, pour over, percolator, etc. Therefore, the invention is not limited to the specific implementations discussed herein.

Two Dimensional Brew Selection Interface

[0047] As noted above, in some embodiments, it may be desirable to utilize a two dimensional brew selection interface to facilitate user interaction with a drip coffee maker. It has been found, in particular, that achieving a specific coffee flavor profile is difficult for novice consumers, as well as difficult on most drip coffee makers, which lack an ability to control a number of the brewing parameters that can affect a coffee flavor profile.

[0048] Industry groups have attempted to develop standards for coffee flavor profiles, and have developed, for example, a coffee brewing control chart that charts, on one axis, strength, which

represents the concentration of solubles in the coffee, and extraction or development, which represents the yield of solubles in the coffee. Strength may be defined in terms of total dissolved solids (TDS), e.g., the percentage (alternatively, the amount) of the end product that includes dissolved solids extracted from the coffee grounds during the brewing process. Extraction or development may be defined in terms of the percentage of the coffee grounds that have been extracted, and thus have moved into the liquid phase. In the standard coffee brewing control chart, strength varies between about 0.80% to 1.60%, and extraction or development varies between about 14% and about 26%. It has also been found that strength and extraction/development are linked by a brewing ratio, representing the ratio of the ground coffee and water used as inputs to the drip coffee brewing process.

[0049] An additional characteristic of the coffee brewing control chart is that various characteristics of the end product may be mapped onto the chart, with strength varying between strong and weak, and with extraction or development varying between under-developed, which tends to produce sour or grassy flavors, and bitter. A “golden cup” standard is generally represented at the center of the chart, where a balance exists between both strong and weak and between under-developed and bitter. It has also been found that certain flavor attributes map to different regions of the chart. Moreover, some flavor attributes may represent “taste-alike” attributes, meaning the flavor attributes mimic the flavors of other foods or beverages, e.g., blueberries, citrus fruits, chocolate, vanilla, etc.

[0050] Utilizing a coffee brewing control chart, however, is generally beyond the capacity of many novices, and controlling the various brewing parameters that may impact the coffee flavor profile can be difficult and tedious. It has been found, for example, that the water temperature, the brewing ratio (e.g., the ratio of the weight or volume of the ground coffee and the weight or volume of the water), and the water/coffee contact time can have a significant impact on the coffee flavor profile, as can the roast level of the coffee beans (i.e., how long and/or thoroughly the coffee beans have been roasted). Moreover, the grind size of the coffee beans can also affect the coffee flavor profile as finer grinding generally exposes a greater overall surface area to the water, and thus increases extraction rate. In some drip coffee brewing operations, “blooming” may also be performed, where the ground coffee is wetted down at the start of the operation to allow carbon dioxide to be released from the ground coffee in the form of bubbles, rather than being absorbed into the coffee.

[0051] It will be appreciated that some of the aforementioned brewing parameters may be directly controllable (e.g., water temperature, grind size), while others may be indirectly controllable (e.g., brewing ratio and water/coffee contact time, and in some instances blooming time). Brewing ratio may be controlled, for example, by controlling the volume or mass of coffee grounds and water used in the operation, while water/coffee contact time and blooming time may be controlled by controlling the rate of water flow into a brew basket, controlling the rate of water flow out of the brew basket, or both.

[0052] Additional brewing parameters that may impact a coffee flavor profile, either directly or indirectly, may include, for example, coffee bean type (e.g., brand, model, variety), water pressure, flow rate, and valve position (where a variable valve provides a varying flow rate). The aforementioned list, however, is not exhaustive, and thus other brewing parameters that at least indirectly affect a coffee flavor profile may be used in other embodiments, as will be appreciated by those of ordinary skill having the benefit of the instant disclosure.

[0053] Given the numerous and often interconnected brewing parameters that are potentially controllable to control a coffee flavor profile, it would be desirable to provide a simplified manner of controlling these various brewing parameters to facilitate a user's selection of a desired coffee flavor profile. Thus, in the illustrated embodiments, a two dimensional brew selection interface is utilized to provide such a simplified manner of control. Brewing parameters may be effectively abstracted out and generated from one or more transfer functions that map inputs to the two dimensional brew selection interface to sets of values of the controllable brewing parameters

utilized in a drip coffee brewing operation.

[0054] A two dimensional brew selection interface, in this regard, may generally be considered to be a graphical user interface control having a plurality of selectable positions distributed in two dimensions. In some instances, for example, a two dimensional brew selection interface may be displayed on a touchscreen such that user input directed to the touchscreen selects a particular position in the two dimensional brew selection interface. In other embodiments, however, other control methodologies may be used, e.g., up/down buttons, knobs, joysticks, sliders, point-and-click (e.g., with a mouse), or any other control methodology that allows for selection of a particular position within a two dimensional area.

[0055] In some embodiments, a two dimensional brew selection interface includes first and second dimensions that are respectively mapped to strength and extraction/development, as those terms are understood within the context of the coffee brewing control chart. A plurality of positions may be distributed across those dimensions, and each position may be mapped to a set of values for a plurality of brewing parameters used when performing a drip coffee brewing operation. As a result, selection of a selected position within the two dimensional brew selection interface selects a set of values for the brewing parameters that are mapped to that position, such that a drip coffee brewing operation may thereafter be performed using the selected set of values.

[0056] Further, in some embodiments, flavor attributes associated with some positions or sets of positions within the two dimensional brew selection interface may be presented to a user in the two dimensional brew selection interface to further assist the user in selecting a desired coffee flavor profile, and in some instances, taste-alike flavor attributes, such as citrus, spice, nutty, fruity, cocoa, honey, herbal, and floral (among others) may be used to identify these flavor attributes. In some embodiments, other information, e.g., the strong/weak and/or under-developed/bitter characteristics mapped to the coffee brewing control chart, may also be presented to a user in the two dimensional brew selection interface to provide the user with another manner of controlling a drip coffee brewing operation. In some embodiments, the mappings between the positions in the two dimensional brew selection interface and the sets of values of the brewing parameters may be determined empirically.

[0057] As will also become more apparent below, a two dimensional brew selection interface may also be usable to display a brew history to a user, with prior drip coffee brewing operations displayed on the two dimensional brew selection interface at locations corresponding to the selected positions used in those operations. In addition, a user may be permitted to rate each drip coffee brewing operation, and have that rating presented on the two dimensional brew selection interface. By doing so, a user may be able to iteratively try different positions, rate the results, and eventually settle on a position that he or she is happy with for future drip coffee brewing operations. Further, in some embodiments, specific positions may be saved as favorites, thereby facilitating recall for future drip coffee brewing operations.

[0058] Now turning to FIG. 3, and with additional reference to FIGS. 4-20, an example operational sequence for controlling a drip coffee brewing operation using a two dimensional brew selection interface is illustrated at **100**. Operational sequence **100** may be executed, for example, by one or more processors in drip coffee maker **10** of FIGS. 1-2, or alternatively, at least some of the operations in operational sequence **100** may be executed by one or more processors of a user device in communication with drip coffee maker **10**, e.g., user device **86** of FIG. 2. It is assumed for the purposes of this example that the two dimensional brew selection interface is displayed on a touchscreen display (either on the drip coffee maker or on the user device), whereby a user may select a position on the two dimensional brew selection interface by touching the touchscreen display at an appropriate position. With other user interfaces and controls, however, other types of user interaction may be used, so the invention is not limited to the touchscreen-based interaction described herein.

[0059] With initial reference to FIG. 4, a home display **150** may initially be displayed to a user, and

may include controls used for selecting a standard drip coffee brew operation (control **152**), a custom drip coffee brew operation (control **154**) or a saved drip coffee brew operation (controls **156**). Additional functions, such as changing settings, configuring the drip coffee maker, etc., may be accessed through a settings control **158**. Assuming, for example, a user selects control **154** to request a custom drip coffee brew operation, block **102** of FIG. **3** may display a two dimensional brew selection interface to the user and wait for user input.

[0060] FIG. **5**, for example, illustrates a custom brew display **170** that includes an example implementation of a two dimensional brew selection interface **172** consistent with some embodiments of the invention. Interface **172** includes a vertical strength axis or dimension **174** and a horizontal development or extraction axis or dimension **176**, and in some embodiments, these dimensions **174**, **176** may generally correspond to the strength and development/extraction axes of the standard coffee brewing control chart. However, rather than utilizing the ranges of percentages corresponding to those in the coffee brewing control chart, it may be desirable to use other ranges and/or units. In the illustrated embodiment, for example, the dimensions are unitless and span ranges of 0-100, thereby providing a 100×100 two dimensional array of selectable positions. It will be appreciated that finer or coarser resolution may be used in other embodiments (e.g., a 10×10 array may be sufficient in some embodiments), and that various other units may also be used. The “golden cup” position of the coffee brewing control chart may also optionally be represented by an icon or indicator **178** in some embodiments.

[0061] Display **170** may include additional controls, including a brew history control **180**, a customize control **182** and an info control **184**, the operation of which will be discussed in greater detail below.

[0062] For the purposes of this example, assume that upon being presented with display **170**, a user selection the position represented at **186**. Returning to FIG. **3**, such a selection passes control to block **104** to display a volume selection display and wait for further user input. FIG. **6**, for example, illustrates a volume selection display **190** that queries the user for the number of cups, or the total volume (e.g., in ounces), he or she wishes to brew. A set of up/down controls **192**, for example, may be used to make the desired volume selection, and a user may select a continue control **194** when the desired volume has been selected.

[0063] In some embodiments, prior to presenting a user with display **170**, the intermediate display **200** of FIG. **7** may be presented, providing the user with information about the selected position (here, a strength of 81 and a development/extraction of 62), a continue control **202** to proceed to display **190**, as well as additional user controls, e.g., a control **204** to provide additional information about the selection, a control **206** to save the selection as a favorite, and a back control **208** to enable the user to go back to display **170** to re-select a position.

[0064] Returning again to FIG. **3**, once the user has made a volume selection, control passes from block **104** to block **106** to calculate the brewing parameters to be used for the drip coffee brewing operation. The brewing parameters may be based, in part, on the selected position in the two dimensional brew selection interface, e.g., using a set of values for the brewing parameters that are mapped to the selected position. The brewing parameters may be scaled, for example, based upon the selected volume, e.g., to determine a suitable volume or weight of coffee grounds necessary to provide a desired brewing ratio based upon the volume selected by the user. In some embodiments, for example, the primary brewing parameters may include water temperature, brewing ratio, water/coffee contact time, and bloom time, and as a result of these primary brewing parameters, various secondary brewing parameters may be calculated, e.g., flow rates, valve positions, etc. suitable for providing the desired water/coffee contact time, as well as the volume or weight of coffee grounds matching the selected brewing ratio and given the volume of water selected by the user. These brewing parameters, however, are not exclusive, so the invention is not limited to this specific combination of brewing parameters.

[0065] Once the brewing parameters have been calculated, control passes to block **108** to determine

whether a grinder is attached to or otherwise connected to the drip coffee maker. If not, the user may be prompted to add the proper amount of coffee grounds (either pre-ground, or ground by an independent coffee grinder) to the brew basket (block **110**). FIG. **8**, for example, illustrates a ground coffee amount display **210** that may be displayed in some embodiments, providing the user with the specific amount of coffee grounds to add to the brew basket, as well as providing a brew control **212** that may be selected once the coffee grounds have been added and the drip coffee brewing operation is ready to be performed.

[0066] Returning to FIG. **3**, and specifically to block **108**, if there is a controllable grinder in communication with the drip coffee maker, control passes to block **112** to activate the grinder to grind the determined amount of coffee grounds. Grinding may occur at a fixed grind size, or if grind size is a controlled brewing parameter, the grinder may be configured to operate at the desired grind size. The amount of coffee grounds may be controlled by controlling a duration of operation of the grinder, or alternatively, based upon weight, e.g., as sensed by scale **52** of FIG. **1**. Thereafter, in block **114**, an add coffee display may be displayed to the user, prompting the user to add the coffee ground by the grinder to the brew basket and select a brew control to continue. In some embodiments, a display similar to display **210** of FIG. **8** may also be displayed in block **114**, potentially omitting the volume of coffee grounds to be added, given that the grinder has already ground the desired amount.

[0067] Upon selection of a brew control in either block **110** or block **114**, control then passes to block **116** to perform the drip coffee brewing operation using the calculated values for the brewing parameters. In some embodiments, a display such as display **220** of FIG. **9** may be displayed during the operation. In addition, in some embodiments it may be desirable to allow for an operation to be paused (e.g., by selecting control **222**), which may result in the operation being paused and a display such as display **230** of FIG. **10** being presented to the user. At that time, a user may be able to resume the operation by selecting control **232**, or alternatively cancel the operation by selecting control **234**.

[0068] As noted above, in some embodiments at least a portion of operational sequence **100** may be performed on a user device. As such, and as represented in block **118** of FIG. **3**, if the operations performed in blocks **100-114** are performed in a user device, rather than performing the drip coffee brewing operation in block **116**, a user device may instead communicate data to a drip coffee maker in block **118** to cause that drip coffee maker to perform the drip coffee brewing operation. The data communicated may include, for example, the calculated values for the brewing parameters, or other data that enables a drip coffee maker to calculate the values itself.

[0069] Regardless of whether the operational sequence is performed by the drip coffee maker or a user device, once the drip coffee brewing operation is complete, either of block **116** or block **118** passes control to block **120** to display a brewing complete display and wait for user input. FIG. **11**, for example, illustrates one suitable brewing complete display **240**, which indicates that the coffee is ready and provides a number of controls that are selectable by a user. A rate this brew control **242**, for example, may be used to rate the coffee that was just brewed, e.g., as represented by block **122** of FIG. **3**. A save to favorites control **244**, for example, may be used to save the coffee that was just brewed as a favorite (e.g., based upon the selected position in the two dimensional brew selection interface), e.g., as represented by block **124** of FIG. **3**. A return to menu control **246** may be used to return to main display **150** of FIG. **4**.

[0070] FIG. **12** illustrates an example rate display **250** that may be presented to a user when rating the coffee in block **122** of FIG. **3**. A user may be presented with a control **252** suitable for rating the coffee, e.g., with 0 to 5 stars, or using other denominations. A return to menu control **254** may also be provided to return to main display **150** of FIG. **4**. When rating the coffee, the rating may be stored along with additional information, e.g., the selected position (e.g., the strength and development values), a timestamp, other brewing parameters (e.g., the selected volume), etc. As discussed below, this information may later be used to provide a user with historical brew

information.

[0071] FIG. 13 illustrates an example save to favorites display 260 that may be presented to a user when saving the coffee as a favorite in block 124 of FIG. 3. A user may be presented with a set of controls 262 that may be overwritten with the current coffee (e.g., the selected position). A return to menu control 264 may also be provided to return to main display 150 of FIG. 4. Saving the coffee as a favorite enables a user to brew using the same brewing parameters when selecting the appropriate control 156 in main display 150.

[0072] Returning to FIG. 3, and specifically to block 100, selection of control 180 in display 170 of FIG. 5 may result in the display of a user's brew history. While brew history may be displayed in a number of manners in other embodiments, in the illustrated embodiment, a display such as display 270 of FIG. 14 is displayed, including a two dimensional brew selection interface 272 overlaid with icons or indicators 274 corresponding to the positions of prior drip coffee brewing operations, i.e., such that the location of each icon or indicator in display 270 generally corresponds to the associated position of the prior drip coffee brewing operation in the two dimensional brew selection interface. The position of each icon or indicator 274 may be based upon the selected position that was used to configure the drip coffee brewing operation, and moreover, it may be desirable to display each icon or indicator 274 as a rating indicator that represents the user's rating of the corresponding drip coffee brewing operation, e.g., using a number corresponding to the number of stars, a numerical rating, a color, a pattern, a size, etc. It will be appreciated that by displaying the brew history in this manner, a user is able to visualize where in the two dimensional brew selection interface the user has brewed coffee, and where the results have been better or worse. In addition, a user is given guidance as to possible positions to select in the future if he or she wants to continue to improve the coffee flavor profile. In FIG. 14, for example, two prior operations have been rated with 5 stars, so a user may wish to select a different position in close proximity to those prior operations (perhaps at a midpoint between those operations) to see if he or she likes the results better than past operations.

[0073] In addition, each icon or indicator 274 is selectable by a user to obtain additional information about the corresponding drip coffee brewing operation and/or to potentially use the same brewing parameters to brew additional coffee. Thus, as illustrated in block 128 of FIG. 3, upon selection of an icon or indicator, information about the selection may be displayed and additional user input may be received. FIG. 15, for example, illustrates a suitable display 280 for displaying additional information about a prior drip coffee brewing operation, including, for example, the selected position (strength and development values) and the rating. At this time, the user may have the option of brewing additional coffee using the same brewing parameters (control 282) or saving to favorites (control 284). Returning again to FIG. 3, if the user does wish to brew additional coffee, block 128 may pass control to block 104 to proceed with the drip coffee brewing operation using the same brewing parameters.

[0074] Returning again to FIG. 3, and specifically to block 100, selection of control 182 in display 170 of FIG. 5 may result in the display of custom brew settings in block 130. FIG. 16, for example, illustrates an example display 290 within which are disclosed various configurable brewing parameters, here water temperature (controllable using up/down arrow controls 292), bloom time (controllable using up/down arrow controls 294) and brewing ratio (controllable using up/down arrow controls 296). Thus, display 290 may be suitable for users with a desire for more granular control over a drip coffee brewing operation. After configuring the brewing parameters, selection of continue control 298 may initiate a drip coffee brewing operation, with block 130 of FIG. 3 passing control to block 104 to proceed with the drip coffee brewing operation using the selected brewing parameters.

[0075] Returning again to FIG. 3, and specifically to block 100, selection of control 184 in display 170 of FIG. 5 may result in the display of a flavor attribute overlay onto a two dimensional brew selection interface (block 132). FIG. 17, for example, illustrates an example display 300 within

which is displayed a two dimensional brew selection interface **302** along with various flavor attribute labels or indicators **304** corresponding to various flavor attributes and positioned generally in regions of the coffee brewing control chart that are associated with such flavor attributes. In some embodiments, each label or indicator **304** may be selectable and may be mapped to a specific position in the two dimensional brew selection interface **302** such that selection of any portion of the label or indicator will result in selection of the same position, despite the fact that the actual position selected is offset from that mapped to the label or indicator. In other embodiments, however, the labels or indicators may merely be informational, and selection of a position in the two dimensional brew selection interface may be independent of any label or indicator. In display **300**, labels or indicators **304** are provided for various taste-alike flavor attributes, including citrus, spice, nutty, fruity, cocoa, floral, herbal, and honey, along with an additional label or indicator for “balanced”, representing the center point or golden cup standard for the two dimensional brew selection interface.

[0076] In the illustrated embodiment, selection of a label or indicator **304** from display **300** causes the display of a selection information display providing descriptive information about the flavor attribute, e.g., as illustrated in block **134** of FIG. 3. FIG. 18 illustrates an example display **310** for the “citrus” flavor attribute, which displays information about the flavor attribute (“this region of the chart enhances the sweet and sour profile of your beans, and your brew will be strong but with low bitterness”). A brew control **312** may be provided to proceed with a drip coffee brewing operation using the selected position corresponding to the selected flavor attribute (whereby control passes from block **134** to block **104** of FIG. 3). A back control **314** may return to display **300** of FIG. 17 to allow for further perusing by the user.

[0077] Returning again to FIG. 3, and specifically to block **100**, it may also be desirable to provide a user with a control chart overlay when desired, e.g., to display additional details about the coffee brewing control chart upon which the two dimensional brew selection interface is based (block **136**). FIG. 19, for example, illustrates an example display **320** within which is displayed a two dimensional brew selection interface **322** along with an overlay of characteristic labels or indicators corresponding to the different regions of the coffee brewing control chart, e.g., strong under-developed, strong, strong bitter, under-developed, balanced, bitter, weak under-developed, weak, and weak bitter. A user may be able to select a position in interface **322** and proceed with a drip coffee brewing operation in a similar manner to that discussed above (by passing control from block **136** to block **104** in FIG. 3). In addition, a back control **324** may return to display **170** of FIG. 5.

[0078] Returning again to FIG. 3, and specifically block **138**, it may also be desirable in some embodiments, particularly with smaller touchscreen displays, to allow for a user to zoom in on a portion of a two dimensional brew selection interface before selecting a position in the interface. Thus, rather than selecting a position in one action, a user may first select one region (e.g., one quadrant) of the interface to cause that region (e.g., quadrant) of the interface to be zoomed in, thereby allowing the user to select a specific position with greater precision. FIG. 20, for example, illustrates a display **330** that may be displayed in some embodiments after a user initially selects the upper right quadrant of two dimensional brew selection interface **170** of FIG. 5, where a quadrant **332** of the two dimensional brew selection interface has been displayed, and a user is able to select a position within that quadrant. Additional functions, e.g., viewing brew history (control **334**), customizing brewing parameters (control **336**) and displaying further information (control **338**) may still be supported.

[0079] Now turning to FIG. 21, as noted above, it may also be desirable to configure a drip coffee brewing operation at least in part based upon the type of coffee beans being used in the operation. While various alternative manners of obtaining information about the type of beans being used may be used, in some embodiments a bar code on the packaging used for the coffee beans may be used to access a remote or local database for use in configuring one or more brewing parameters for a

drip coffee brewing operation. FIG. 21 in particular illustrates an operational sequence 350 that begins in block 352 by capturing a bar code of the packaged coffee beans. The bar code may be captured, for example, using scanner 96 of FIG. 2, or alternatively, by a camera of a user device. Once the bar code is scanned, and a SKU or other product identifier is determined, a remote service may be accessed (block 354) to obtain data associated with the coffee beans. The data may include information such as roast level, variety, brand, model, etc., or may include one or more brewing parameters to be used when brewing with the beans, scaling factors that may be used based upon properties of the beans, suggested grind sizes, or even specific brewing recipes, e.g., as created by producers or third parties. The data may be stored for future use (block 356) and then at a later point, one or more drip coffee brewing operations may be performed using the stored coffee bean data (block 358).

Brewing Parameter Suggestions

[0080] It may also be desirable in some embodiments to provide brewing parameter suggestions to a user based, at least in part, on user feedback and/or history. In particular, in some embodiments, users may be able to brew coffee specifically tailored to their personal preferences based upon a number of system inputs to generate user-specific suggested brewing parameters. In addition, with the use of the herein-described two dimensional brew selection interface, these suggested brewing parameters may be presented to users in a simple and clear manner that removes much of the guesswork for users. Furthermore, repeated usage of brewing parameter suggestions may optimize the values of multiple brewing parameters used in a coffee brewing operation in an iterative and intuitive manner.

[0081] In some embodiments, for example, a set of values of a plurality of brewing parameters may be “favorited” by a user and saved for future coffee brewing operations. This “favorite” may then be subsequently modified over time as user ratings are accumulated after subsequent coffee brewing operations are conducted. Each additional suggestion with a user rating may modify a suggested set of values for the plurality of brewing parameters based upon both the sets of values used for the prior coffee brewing operations and the user ratings assigned to those prior coffee brewing operations, thereby adjusting the suggested set of values to progressively approach an ideal or optimum set of values.

[0082] Moreover, when used with a two dimensional brew selection interface as described herein, this iterative approach may be visually apparent to a user, as the position of a suggested indicator associated with a suggested set of values may be displayed in a position of the two dimensional brew selection interface along with additional indicators, e.g., a favorite indicator associated with the current favorite and displayed at a position corresponding to the set of values associated with the current favorite and/or one or more historical indicators associated with one or more prior coffee brewing operations and displayed at positions corresponding to the sets of values associated with those prior coffee brewing operations.

[0083] In addition, as will become more apparent below, brewing parameter suggestions, as well as favorites developed from such brewing parameter suggestions, may be tailored to particular coffee brewing operations, coffee bean types, coffee drink types, etc. For example, a user may have different brewing preferences for different bean types, such as different roast levels, origins, varieties, etc., different brewing processes and/or drink types, such as coffees, espressos, lattes, cappuccinos, coffee with sugar, coffee with syrup, etc. As such, it may be desirable in some embodiments to maintain different data sets or profiles for different bean types, brewing processes, and/or drink types (including for different additives such as sugar, milk, syrup, etc.). In some embodiments, a user may be presented with a rating history and receive suggestions specific to these subsets by selecting a bean and/or drink profile. Further, in some embodiments, a user may be presented with one or more prompts in order to ascertain the drink type to be brewed, e.g., “will you add cream/sugar/milk/syrup?”

[0084] Brewing parameter suggestions in some embodiments may also be based upon aggregated

feedback data from multiple users. In addition, in some embodiments, additional data, e.g., user profile data, time, date and/or location data, etc., may be used, e.g., to tailor brewing parameter suggestions to particular times of the day (morning vs. evening), times of the year (winter vs. summer), or geographical areas. It will be appreciated, for example, that different geographical areas often have different coffee cultures and taste preferences, so in some embodiments, users' personal brewing parameter suggestions may be weighted or otherwise based in part on the preferences of other users in their respective geographical areas.

[0085] An advantage of the herein-described approach is that a user may be guided towards an optimum set of values for his or her personal taste preferences simply by brewing coffee using brewing parameter suggestions and rating the brewed coffee. Doing so enables users without a deep understanding of the complex interactions between different brewing parameters to brew coffee that matches their personal taste preferences.

[0086] As will become more apparent below, various algorithms and approaches may be utilized to generating brewing parameter suggestions in various embodiments. In some embodiments, for example, a mathematical distance or other optimization algorithm such as an n-dimensional weighted Euclidian distance algorithm may be employed to determine an optimized position in a Euclidian space based upon both the ratings and the positions of a plurality of prior coffee brewing operations. In some embodiments, the n dimensions may correspond to various brewing parameters or combinations of brewing parameters. For example, in some embodiments, sets of brewing parameters may be mapped to the two dimensional space utilized by the two dimensional brew selection interface, e.g., with axes corresponding to strength and development/extraction, such that the positions of the prior coffee brewing operations and a brewing parameter suggestion generated therefrom may be represented with corresponding indicators in the two dimensional brew selection interface.

[0087] In other embodiments, various artificial intelligence or machine learning techniques may be utilized to generate brewing parameter suggestions based on any or all of the aforementioned data.

[0088] In still other embodiments, as noted above, rather than utilizing a two dimensional space as a multi-dimensional space, a multi-dimensional space with three or more dimensions may be supported, e.g., to consider factors beyond strength and development/extraction such as drinking temperature, bean roast conditions, etc. It should be noted that such a space may be presented in a two dimensional user interface in some embodiments (e.g., through visualization of a 3D cube on a two dimensional display), or otherwise in another N-dimensional user interface that may or may not be the same number of dimensions as the multi-dimensional space. In some embodiments, for example, the number of dimensions for brewing parameter suggestions may differ from the number of dimensions in the user interface such that the position of the suggestion is not necessarily presented to a user at a specific n-dimensional point in the user interface corresponding to the multi-dimensional space.

[0089] In addition, the manner in which prior coffee brewing operations may be rated in order to generate a brewing parameter suggestion may vary in different embodiments. As noted above, ratings may be based on numerical values (e.g., 0 to 5 stars), or alternatively based on various relative schemes (e.g., good, bad, ok) that may be mapped to numerical values. In some instances, the simple fact that a particular prior coffee brewing operation was favorited by a user may be utilized as an indicator of rating, as the act of favoriting can be considered to be an indication of whether or not a user enjoyed a prior coffee brewing operation. In other embodiments, ratings may be associated with more general flavors or user perceptions, e.g., based on one or more prompts presented to a user after the user has tasted coffee. For example, a user may be prompted to select one or more flavor attributes perceived by the user, e.g., taste-alike flavor attributes, or alternatively prompted with one or more queries about the user's perception of the coffee (e.g., "was the coffee too bitter?" or "did you think the coffee was too strong or too weak?"). Ratings in some embodiments therefore may be represented using combinations of values for different categories

rather than being represented using single numerical values.

[0090] Now turning to FIG. 22, this figure illustrates an example operational sequence **400** capable of being performed by drip coffee maker **10** (or another type of coffee maker), or alternatively, by user device **86** or a combination of drip coffee maker **10** and a user device **86**, to generate a coffee brewing suggestion and perform a suggested coffee brewing operation based on the coffee brewing suggestion. In block **402**, a suggested coffee brewing operation is determined, at least in part based upon a set of prior coffee brewing operations (the details of which may be performed in any of the manners discussed above, as well as discussed below in connection with FIGS. 24-27). In the illustrated embodiment, the prior and suggested coffee brewing operations may be associated with corresponding sets of values for a plurality of brewing parameters that are used to control the coffee brewing process performed by the coffee maker. Thus, the suggested coffee brewing operation may be considered to define a suggested set of values to be used when performing the suggested coffee brewing operation.

[0091] Next, in block **404**, a two dimensional brew selection interface may be presented to a user, along with one or more historical indicators (e.g., rating and/or favorite indicators) associated with one or more prior coffee brewing operations and a suggested indicator associated with the determined suggested coffee brewing operation. In some embodiments, for example, the two dimensional brew selection interface may be similar to two dimensional brew selection interface **172** of FIG. 5, where strength is used for the vertical (y) axis or dimension and development or extraction is used for the horizontal (x) axis or dimension, and where the sets of values for the suggested and prior coffee brewing operations map to specific positions along each of the axes. As such, the suggested indicator may be presented at a position in the two dimensional brew selection interface that allows a user to visualize it in the context of the historical indicators for the prior coffee brewing operations.

[0092] Next, in block **406**, a user selection of the suggested coffee brewing operation may be received, e.g., via a touch or other suitable user input directed to the suggested indicator or another user control, via voice or gesture input, etc., and in block **408**, the set of values associated with the suggested coffee brewing operation may be applied to initiate a suggested coffee brewing operation. When, for example, sequence **400** is implemented on a coffee maker, block **406** may include performing the suggested coffee brewing operation with the coffee maker using values from the set of values, while when sequence **400** is implemented on a user device, block **406** may include communicating a signal to the coffee maker to cause the coffee maker to perform the suggested coffee brewing operation with the coffee maker using values from the set of values. In addition, it will be appreciated that in some embodiments, additional user input and/or actions may be necessitated, e.g., to place suggested coffee beans in a grinder, grind a suggested amount of coffee beans (and in some instances, using a suggested grind size) and/or otherwise appropriately set up the coffee maker to brew (e.g., by placing ground coffee and a filter into the brew basket). Where, for example, a particular brewing parameter must be manually configured by the user in order to utilize the suggested value for that brewing parameter, user input and/or action will generally be needed prior to starting the actual brewing process.

[0093] In still other embodiments, e.g., where the suggested coffee brewing operation is performed using a manual brewing process such as pour over, French press, cold brew, etc., initiation of a suggested coffee brewing operation may include instructing a user of various brewing parameters to be used in the manual brewing process. For a pour over brewing process, for example, a user may be presented with brewing parameters such as bloom time, water temperature, quantity and grind size of coffee grounds, and various quantities of water and points in time at which to add those quantities to a brew container.

[0094] Once brewing is complete, a user's rating of the coffee brewing operation may be received in block **410**, and where a favorite coffee brewing operation is tracked, the favorite may be updated to match the suggested coffee brewing operation, such that the position of the favorite indicator is

thereafter updated to that of the suggested indicator.

[0095] FIG. 23 next illustrates an example two dimensional brew selection interface **420** incorporated into a display **422**, e.g., the display of a coffee maker or a user device, and suitable for displaying a suggested indicator for a suggested coffee brewing operation. Two dimensional brew selection interface **420** may include a plurality of positions distributed across first and second dimensions (e.g., strength in a vertical (y) axis or dimension and development or extraction in a horizontal (x) axis or dimension), and each position may be mapped to a set of values for a plurality of brewing parameters used when performing a coffee brewing operation.

[0096] A set of historical indicators **424**, **426**, **428** may be overlaid on the interface at positions corresponding to sets of values of the plurality of brewing parameters used for their associated prior coffee brewing operations. Historical indicator **424** is a rating indicator that, in addition to indicating the position of the associated prior coffee brewing operation, also indicates a user's rating of that prior coffee brewing operation. In the illustrated embodiment, for example, the rating is represented using the size of the rating indicator, with larger sizes indicating higher ratings, although other display characteristics, e.g., colors, patterns, icons, etc., may be used to represent different ratings in other embodiments.

[0097] Historical indicator **426**, which is represented by a star or other suitable icon, is a favorite indicator, which indicates that the user has favorited the associated prior coffee brewing operation. In the illustrated embodiment, multiple favorite indicators may be displayed, with the most recently used, illustrated at **428**, being used to represent a “current” favorite for the user. In some embodiments, the current favorite indicator **428** may be displayed differently from other favorite indicators **426** to allow a user to distinguish their current favorite from past favorites. In other embodiments, however, only a single favorite may be tracked, such that, for example, other favorite indicators **426** are visually represented similar to rating indicators **424**. Displaying current and past favorites concurrently, however, may in some instances provide a user with greater context for ascertaining how they have progressed towards their optimum brew parameter settings.

[0098] Additional information, including, for example, flavor attribute labels or indicators **430** or any of the other additional information described above, may also be displayed on two dimensional brew selection interface **420**, as will be appreciated by those of ordinary skill having the benefit of the instant disclosure.

[0099] In addition, a suggested indicator **432**, representing the position corresponding to a determined suggested coffee brewing operation, may also be displayed, and in some instances, an additional indicator or animation (e.g., arrow **434**) may be displayed to highlight the transition within the two dimensional space from the current favorite coffee brewing operation to the suggested coffee brewing operation. In some embodiments, selection of a suggested coffee brewing operation may be performed by selecting suggested indicator **432**, while in other embodiments, selection may be performed by selecting another user control such as button **436**.

[0100] Additional controls, e.g., a toggle button **438** that activates or deactivates the suggestion functionality, may be provided in association with two dimensional brew selection interface **420** in some embodiments. Other interfaces and controls may be used in other embodiments.

[0101] FIG. 24 next illustrates an example embodiment of a suggestion determination system **450**, which may be used, for example, to determine a suggested coffee brewing operation in block **402** of FIG. 22. System **450** may include a suggestion determination module **452** that may receive various types of input. For example, in some embodiments module **452** may receive one or more prior coffee brewing operations performed for the user, along with associated ratings (block **454**), and may output a suggested coffee brewing operation as illustrated in block **456**.

[0102] In other embodiments, however, additional data may be used to generate a suggested coffee brewing operation. For example, module **452** may receive one or more prior coffee brewing operations performed for other users (either on the same coffee maker or for different coffee makers), along with associated ratings (block **456**), such that the determined suggestion may be

based on the aggregated ratings of multiple users. Module **452** may also receive time, date and/or location data (block **460**), which may be used, for example, to tailor a brewing parameter suggestion towards a particular time of day, date (e.g., time of the year) and/or geographical area. [0103] User profile data (block **462**) may also be provided to module **452**, e.g., to reflect any user preferences established in their profile. For example, a user may specify in his or her profile a preference for particular flavors, roast levels, drink types, additives used, etc., such that suggestion determinations may be based in part on the user's profile.

[0104] Coffee bean data (block **464**), e.g., origin, variety, roast level, etc., may also be provided in some embodiments, thereby enabling a suggestion determination to be tailored to a particular coffee bean to be used in the brewing process. The coffee bean data may be manually input by a user in some embodiments, or may be determined automatically and/or programmatically, e.g., by scanning a barcode as discussed above in connection with FIG. **21**, by analyzing coffee beans to access roast level, etc. Coffee bean data may also be associated with prior coffee brewing operations such that suggestion determinations may be weighted to favor prior coffee brewing operations performed with the same or similar coffee beans.

[0105] Moreover, drink type data (block **466**) may be provided in some embodiments, e.g., via user selection or prompting of a user, to assist with tailoring a suggestion determination to the type of drink within which the coffee to be brewed is to be used. The drink type, for example, may be based on the additive(s) used, e.g., sugar, creamer, milk, syrup, cocoa, spices, etc., the brewing process used, e.g., cold brew, espresso, drip coffee, pour over, French press, percolation, etc., or any other coffee-based drink variation that could potentially be affected by the taste profile of the brewed coffee used in the drink. Drink type data may also be associated with prior coffee brewing operations such that suggestion determinations may be weighted to favor prior coffee brewing operations performed for similar types of coffee-based drinks.

[0106] Based on any or all of the data received from blocks **454-466**, module **452** may generate the suggested coffee brewing operation in block **456**. In addition, as illustrated in block **468**, module **452** may also generate, as one or more of the brewing parameters for the suggested coffee brewing operation, a coffee bean recommendation for the user. Module **452**, for example, may be able to ascertain from a user's history and ratings where a user prefers a lighter or darker roast, a particular variety or blend of varieties (e.g., *robusta*, *arabica*, etc.), or a particular origin (e.g., southeast Asia vs. South America). Based upon these preferences, module **452** may also be configured to recommend a particular coffee bean matching the user's preferences, although in other embodiments, module **452** may only provide a user with generalized recommendations.

[0107] FIG. **25**, for example, illustrates a display **470** providing a coffee bean recommendation **472**, which recommends a dark roast **60/40** blend from Sumatra. A user control **474** may be used to dismiss display **470** or proceed with a coffee brewing operation, while in some embodiments, a user may be provided with a user interface to order or purchase coffee beans matching the recommendation, e.g., via a control **476**.

[0108] Returning to FIG. **24**, module **452**, in some embodiments, may utilize a mathematical distance optimization algorithm based on equations (1)-(14) below:

$$[00001] F(F_x, F_y) \quad (1) \quad F'(x, y) = F(x, y) + S(x, y) \quad (2) \quad H_i(x, y) = R_i \quad (3)$$

$$P_i = d(H_i(x, y), F) \quad (4) \quad R_F \equiv H_i(F) \quad (5) \quad D_i = R_F - H_i \quad (6) \quad C_i = \frac{D_i}{P_i} \quad (7)$$

$$= \text{preferential scaling factor} \quad (8) \quad = \text{proximal scaling factor} \quad (9)$$

$$X_i = \cos(\tan^{-1}(H_{ix} - F_x)) \quad (10) \quad Y_i = \sin(\tan^{-1}(H_{iy} - F_y)) \quad (11) \quad S_x = .\text{Math.}_i C_i X_i \quad (12)$$

$$S_y = .\text{Math.}_i C_i Y_i \quad (13) \quad F'(x, y) = ((F_x + S_x), (F_y + S_y)) \quad (14)$$

[0109] In particular, a favorite, F, may have an (x, y) position on a map (F.sub.x, F.sub.y) corresponding to a two dimensional space. A new position F' in the two dimensional space may be

suggested by the algorithm by adding a suggestion term $S(x, y)$ to the original position. The S term may be found by calculating a modification coefficient $C_{sub.i}$ for each rated point in a historical data set H . Each historical data point H_i has rating $R_{sub.i}$, and the coefficient may be determined from the proximal distance ($P_{sub.i}$) of $R_{sub.i}$ from F , and the difference ($D_{sub.i}$) between the user rating of that point and rating of the Favorite. Scaling factors for each are chosen to scale the relative effect of rating (α) and proximity (β) on the suggestion algorithm. The x and y components of the suggestion may be found independently using the cartesian angle between each historical rating and the Favorite position. The actual suggestion may then be determined as a superposition of independent directional effects. Additional effects based on other relevant user data can be added to the suggestion term(S) in some embodiments to consider additional factors such as bean type, milk preference, desired drinking temperature, ratings of similar users, etc.

[0110] Other manners of implementing module **452** may be used in other embodiments. For example, in some embodiments, a machine learning model trained on a plurality of historical coffee brewing operations, and optionally, based on any of the other data discussed above in connection with FIG. **24**, may be used to generate brewing parameter suggestions consistent with the invention.

[0111] FIG. **26**, for example, illustrates one example manner of implementing a coffee brewing suggestion system **480** capable of receiving input data **482** and processing that data with a trained coffee brewing suggestion machine learning model **484** to generate brewing parameter suggestions. In some embodiments, for example, model **484** may be implemented as a deep neural network (DNN) including an input layer **486**, one or more intermediate layers **488**, and an output layer **490** including one or more brewing parameter suggestion heads **492** and one or more coffee bean suggestion heads **494**. In some embodiments, for example, one or more intermediate layers **488** may include one or more convolutional layers. The dimensions/shape of input layer **486** may be dependent on the shape of the image data to be applied, while the dimensions/shape of each output head **492**, **494** may be dependent on various factors such as how many class probabilities are to be predicted, among others. In some embodiments, multiple convolution layers may be provided, and max pooling and/or other layers such as affine layers, softmax layers and/or fully connected layers may optionally be interposed between one or more of the convolution layers and/or between a convolution layer and the output layer. Other embodiments may not include any convolution layer and/or not include any max pooling layers, and in still other embodiments, other machine learning models may be used, e.g., Bayesian models, random forest models, Markov models, etc.

[0112] FIG. **27** illustrates a system for training model **484**, e.g., using a training engine **500** that utilizes training instances **502** generated by a training instance generator **504** from a set of input data **506**. Generator **504** may process input data **506** to generate, for each training instance **502**, an input **508** including input data, and an output **510** including brewing parameter and/or coffee bean classification. In training model **484**, training engine **500** may apply the training instance input **508** to model **484** and process the training instance input **508**, utilizing model **484** and based on current parameters of model **484**, to generate an output having a dimension that conforms to the dimension of training instance output **510**. Training engine **500** may then compare the generated output to the training instance output **510**, and update one or more parameters of model **484** based on the comparison. For example, training engine **500** may generate an error based on differences between the generated output and the training instance output **510**, and backpropagate a loss (that is based on the error) over model **484** to update model **484**.

[0113] As model **484** is a multi-head model that incorporates at least one brewing parameter suggestion head **492** and at least one coffee bean suggestion head **494**, different subsets of training instances may be used, thereby co-training the different output heads **492**, **494** and jointly optimizing the model for both types of suggestions. Where coffee bean suggestion is not incorporated into model **484**, however, only training instances associated with brewing parameter suggestion may be used.

[0114] It will be appreciated that, while certain features may be discussed herein in connection with certain embodiments and/or in connection with certain figures, unless expressly stated to the contrary, such features generally may be incorporated into any of the embodiments discussed and illustrated herein. Moreover, features that are disclosed as being combined in some embodiments may generally be implemented separately in other embodiments, and features that are disclosed as being implemented separately in some embodiments may be combined in other embodiments, so the fact that a particular feature is discussed in the context of one embodiment but not another should not be construed as an admission that those two embodiments are mutually exclusive of one another. Various additional modifications may be made to the illustrated embodiments consistent with the invention. Therefore, the invention lies in the claims hereinafter appended.

Claims

1. An apparatus comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by: presenting a two dimensional brew selection interface to a user, the two dimensional brew selection interface including a plurality of positions distributed across first and second dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation; presenting on the two dimensional brew selection interface a plurality of historical indicators respectively associated with a plurality of prior coffee brewing operations, each of the plurality of historical indicators having a respective position in the two dimensional brew selection interface corresponding to a respective set of values for the plurality of brewing parameters used when performing the associated prior coffee brewing operation; determining, based on at least a portion of the plurality of prior coffee brewing operations, a suggested set of values for the plurality of brewing parameters; and presenting on the two dimensional brew selection interface a suggested indicator associated with a suggested coffee brewing operation to be performed, the suggested indicator having a position in the two dimensional brew selection interface corresponding to the suggested set of values for the plurality of brewing parameters.
2. The apparatus of claim 1, wherein a first historical indicator of the plurality of historical indicators is a rating indicator that indicates a user rating associated with a first prior coffee brewing operation of the plurality of prior coffee brewing operations.
3. The apparatus of claim 2, wherein the one or more processors are configured to present the first historical indicator with a size that is based at least in part on the user rating associated with the first prior coffee brewing operation.
4. The apparatus of claim 2, wherein the one or more processors are configured to determine the suggested set of values based at least in part on the user rating of the first prior coffee brewing operation.
5. The apparatus of claim 1, wherein a first historical indicator of the plurality of historical indicators is a favorite indicator that indicates that a user favorited an associated first prior coffee brewing operation of the plurality of prior coffee brewing operations.
6. The apparatus of claim 5, wherein a second historical indicator of the plurality of historical indicators is a rating indicator that indicates a user rating associated with a second prior coffee brewing operation of the plurality of prior coffee brewing operations.
7. The apparatus of claim 5, wherein the one or more processors are configured to determine the suggested set of values based at least in part on the user favoriting the first prior coffee brewing operation.
8. The apparatus of claim 5, wherein the one or more processors are configured to update the position of the favorite indicator to that of the suggested indicator in response to user input received

subsequent to a coffee brewing operation performed using the suggested set of values.

9. The apparatus of claim 1, wherein the one or more processors are further configured to: receive user input from the user selecting the suggested coffee brewing operation; and configure the plurality of brewing parameters for the coffee brewing operation to use the suggested set of values for the suggested coffee brewing operation.

10. The apparatus of claim 9, wherein the apparatus is a mobile computing device, the one or more processors present the two dimensional brew selection interface on and receive the user input from a touch screen display of the mobile computing device, and the one or more processors configure the plurality of brewing parameters for the coffee brewing operation by communicating data to a coffee maker to cause the coffee maker to use the suggested set of values corresponding to the suggested coffee brewing operation.

11. The apparatus of claim 9, wherein the apparatus is a drip coffee maker including a brew basket, a heating element configured to heat water supplied to the brew basket, and a flow control device configured to control water flow through the brew basket, the one or more processors present the two dimensional brew selection interface on and receive the user input from a touch screen display of the drip coffee maker, and the one or more processors further configure the coffee brewing operation by controlling at least one of the heating element and the flow control device during the coffee brewing operation based upon at least a portion of the values in the suggested set of values corresponding to the suggested coffee brewing operation.

12. The apparatus of claim 9, wherein the plurality of brewing parameters includes a water temperature, a water pressure, a water flow rate, a valve position, a water amount, a coffee amount, a grind size, a bean type, a roast level, a brewing ratio, and/or a water/coffee contact time.

13. The apparatus of claim 9, wherein the plurality of brewing parameters includes a coffee amount parameter, and the one or more processors are configured to calculate a value of the coffee amount parameter based at least in part on the position of the suggested indicator in the two dimensional brew selection interface and prompt the user to add an amount of coffee associated with the calculated value of the coffee amount parameter.

14. The apparatus of claim 9, wherein the plurality of brewing parameters includes a coffee amount parameter, and the one or more processors are configured to calculate a value of the coffee amount parameter based at least in part on the position of the suggested indicator in the two dimensional brew selection interface and control a coffee grinder to grind an amount of coffee associated with the calculated value of the coffee amount parameter.

15. The apparatus of claim 9, wherein the user input is first user input, and the one or more processors are further configured to rate the coffee brewing operation after completion of the coffee brewing operation in response to second user input.

16. The apparatus of claim 1, wherein the first dimension corresponds to strength and the second dimension corresponds to development, and the strength is based upon concentration of solubles and the development is based upon yield of solubles.

17. The apparatus of claim 1, wherein the plurality of prior coffee brewing operations is a first plurality of prior coffee brewing operations, and the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters further based on: user profile data associated with a user; a second plurality of prior coffee brewing operations performed on a plurality of coffee makers by a plurality of users; time, date, or location data; coffee bean data; or coffee drink type data.

18. The apparatus of claim 1, wherein the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters using a machine learning model trained based upon a plurality of historical brewing operations.

19. The apparatus of claim 1, wherein the one or more processors are configured to determine the suggested set of values for the plurality of brewing parameters using a mathematical distance optimization algorithm that optimizes the position of the suggested indicator in the two

dimensional brew selection interface based on at least a portion of the plurality of prior coffee brewing operations.

20. The apparatus of claim 19, wherein the mathematical distance optimization algorithm is configured to determine the optimized position of the suggested indicator in the two dimensional brew selection interface using the respective positions of at least a portion of the plurality of historical indicators.

21. The apparatus of claim 1, wherein the one or more processors are further configured to: determine a coffee bean recommendation based on at least a portion of the plurality of prior coffee brewing operations; and present the coffee bean recommendation to the user.

22. An apparatus comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by: presenting an N-dimensional brew selection interface to a user, the N-dimensional brew selection interface including a plurality of positions distributed across N dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation; presenting on the N-dimensional brew selection interface a plurality of historical indicators respectively associated with a plurality of prior coffee brewing operations, each of the plurality of historical indicators having a respective position in the N-dimensional brew selection interface corresponding to a respective set of values for the plurality of brewing parameters used when performing the associated prior coffee brewing operation; determining, based on at least a portion of the plurality of prior coffee brewing operations, a suggested set of values for the plurality of brewing parameters; and presenting on the N-dimensional brew selection interface a suggested indicator associated with a suggested coffee brewing operation to be performed, the suggested indicator having a position in the N-dimensional brew selection interface corresponding to the suggested set of values for the plurality of brewing parameters.

23. An apparatus comprising one or more processors and memory operably coupled with the one or more processors, wherein the memory stores instructions that, in response to execution of the instructions by the one or more processors, cause the one or more processors to suggest a coffee brewing operation by: representing a plurality of prior coffee brewing operations in an N-dimensional space, the N-dimensional space including a plurality of positions distributed across N dimensions, with each position mapped to a set of values for a plurality of brewing parameters used when performing the coffee brewing operation, and with each of the prior coffee brewing operations having an associated user-assigned rating; determining, based on at least a portion of the plurality of prior coffee brewing operations and the user-assigned ratings associated therewith, a suggested position in the N-dimensional space; and suggesting a suggested coffee brewing operation to be performed using the set of values for the plurality of brewing parameters corresponding to the suggested position in the N-dimensional space.
